

Lab 8 Report:

FINAL PROJECT STRATEGIC PLANNING

Lab 8 Instruction:

https://canvas.uw.edu/files/106242897/download?download_frd=1

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Dataset for your project: Real Waste

<https://archive.ics.uci.edu/dataset/908/realwaste>

Feel free to delete or modify pre-written markdown cells below the line

```
In [1]: import os
        from PIL import Image
        import numpy as np
```

Understanding the Data

```
In [ ]: #a) check the size of the database
        # The dataset contains 4752 524xx524 images
        # The dataset is contained in a folder called RealWaste, which has 9 subfolders
        # Carboard, Food Organics, Glass, Metal, Miscellaneous Trash, Paper, Plastic, Textiles, and Wood
        # each of these subfolders contains images of object in there respective categories
        # check the size of the dataset on the disk
        folder_path = 'RealWaste'
        folder_size = get_folder_size(folder_path)
        print(f"Size of the dataset folder '{folder_path}': {folder_size / (1024 * 1024)} GB")

        #b)The dataset contains does not contain any simulated images
        # Check if any images in the dataset are corrupted
        def check_images(folder_path):
            corrupted_images = []
            for root, _, files in os.walk(folder_path):
                for file in files:
                    if file.endswith(('png', 'jpg', 'jpeg')):
                        file_path = os.path.join(root, file)
                        try:
                            img = Image.open(file_path)
                            img.verify() # Verify that it is an image
                        except (IOError, SyntaxError) as e:
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        corrupted_images.append(file_path)
    return corrupted_images
corrupted_images = check_images(folder_path)
if corrupted_images:
    print(f"Corrupted images found: {len(corrupted_images)}")
    for img in corrupted_images:
        print(img)
else:
    print("No corrupted images found in the dataset.")

#c) The image splits are as follows:
#Cardboard: 461
#Food Organics: 411
#Glass: 420
#Metal: 790
#Miscellaneous Trash: 495
#Paper: 500
#Plastic: 921
#Textile Trash: 318
#Vegetation: 436
#the images are all one main feautre in the center and all 254x254

```

Size of the dataset folder 'RealWaste': 656.43 MB

No corrupted images found in the dataset.

Understanding the Task

```

In [ ]: # a) I want am trying to implent classifation and localization of different

# b) Material waste is the worst it has ever been, and it is a major problem
# 220 million tons of plastic waste was produced in 2024 and that number is
# waste such as plastic and metal is not biodegradable and essentially "pois
# I am trying to implent that will help with identification and potentially

#c) For classification we will test if the model labeled the waste correctly
# For localization we will use a bounding box with the format (x, y, width,
# we will then comapre the bounding box the model outputs to that of the tar
# evaluation will be done using the mean average precision (mAP) metric for
# it will calculate the average precision for each class and then take the m
# it also takes into account the intersection over union (IoU) of the predic
# the IoU is calculated by taking the area of the intersection of the two bo

#d) we are looking to achieve a mAP of 0.5 or higher for the classification

#e) we will compare our results to the YOLOv5 model (our model will not be a

```

Developing an initial plan for your project

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In [ ]: #a) What will be the inputs and outputs (including their dimensions) of your
#network model? • What data preparation methods do you plan to use?
# The inputs will be the images, and the outputs will be the class labels ar
# The images will be normalized to have pixel values between 0 and 1
# The class labels will be one-hot encoded, and the bounding boxes will be r

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# The data will be split into training, validation, and test sets using a 70/30 split
# we will also have to annotate the images with the class labels and bounding boxes

#b) I plan to use a convolutional neural network (CNN) for the classification task

#c) I will use CrossEntropyLoss for the classification task and SmoothL1Loss for the regression task

#d) I will evaluate the model as I described above using the mAP metric for the classification task
```